

BUILD FOR BHARAT 2.0 · SAS CU HACKATHON

NEXUS DELTA

Evidence-based capability intelligence for the future workforce

Technical Approach Note

THE NUMBER THAT SHAPED THE DESIGN

0 — the number of employee IDs shared by the two people datasets ($JDS \cap SDS$).

Any analysis that joins these files invents a link. NEXUS DELTA refuses to. It keeps every dataset separate and connects them through a versioned capability taxonomy, then asks three questions:

Know what to build. Which capability should an organisation develop next?

Know what to learn. Which capability should a student or professional develop next?

Know where you can thrive. Will the way a role works fit the way you work?

Team DSA · Chandigarh University

Anwesha Kar · Darshna Parihar · Saamyaraj Baidya

8 executed Python notebooks · SQL · SAS mirror · offline Streamlit application (171 automated checks)

Underlined orange text is clickable: section links jump within this document and notebook links open the executed code in the [project repository](#). The QR code opens the same repository.



The 60-second version

The whole study on one page: what NEXUS DELTA found and why it can be trusted.

Table 1. The decision in one table (all values from results.json).

Capability	Internal evidence	Market evidence	Headroom	Decision
Maths & Statistics	δ 0.55	OR 1.27	59%	PRIORITIZE
AI & ML	δ 0.36	OR 1.38	51%	PRIORITIZE
Coding	δ 0.49	OR 1.52	55%	PRIORITIZE
Dashboards & Storytelling	δ 0.59	OR 0.75	42%	INVESTIGATE
Big Data	δ 0.12	OR 1.09	82%	MAINTAIN

- **Top priority: Maths & Statistics.** Strong inside the company (Cliff's $\delta = 0.55$), a market premium in all 6 models, 59% of juniors still below the ceiling, and the largest modelled gain. It beats the runner-up (AI & ML) in **95.7% of 1,000 bootstrap refits**, ranks first among the four funded capabilities in **91.3%** (NB08), and is first in all 9 sensitivity settings.
- **The paradox: Storytelling.** It is the strongest internal separator ($\delta = 0.59$) yet carries a market *discount* (OR 0.69–0.80 in all six models). Splitting the keyword family shows the discount sits with MIS/reporting keywords (OR 0.66); visualisation keywords alone are neutral (OR 0.98). Verdict: **INVESTIGATE** (fund with reframe) — develop analytical communication, not report production.
- **The trap: Big Data.** The most headroom (82% of juniors below ceiling) but no significant univariate internal link ($\delta = 0.12$, CI crosses 0), and its market premium disappears once experience is controlled. Headroom alone is not a reason to invest: **MAINTAIN**, and revisit with more data.
- **For students:** a posting's title and listed skills predict the exact salary band 44% of the time and land within one band 85% of the time (6 bands; majority-class baseline 24%) — so NEXUS DELTA shows a distribution, never an exact salary.
- **For Bharat:** Tier-2/3 cities account for 9.2% of all cleaned postings (6.7% of data-role postings), and only 29% of their data-role postings reach 10 LPA versus 51% in Tier-1.

Navigate this note

Section	What it answers
1. Problem definition	What decision is being made, by whom, and how success is measured
2. Approach	Why three separate signals, and why no row-level join
3. Data exploration	Every data issue found, its size, and the fix
4. Data analysis	Tests, models, validation and the Storytelling paradox
5. Results & conclusions	The decision, objectives met, hypotheses tested
6. Implications	What changes for each stakeholder, and the product
Appendix	Code, full tables, screenshots, SAS mirror, glossary

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1. Problem Definition & Analytics Objective

IN ONE LINE

Capability investment decisions are usually made on opinion. NEXUS DELTA replaces the opinion with three independent pieces of evidence — and says so when they disagree.

1.1 Context

The SAS CU Hackathon brief frames the problem around data-science jobs, the skills of data scientists and their personality traits, with the objective “to bring out the best information out of available data”. Four files were supplied: **Analytics Jobs** (15,841 postings), **Data Science Jobs** (1,602 rows of company-level openings and salaries), **JDS Skill Traits** (139 junior data scientists, skills on a 1–5 scale and a high/low salary-hike outcome) and **SDS Personality Traits** (161 records, 152 unique IDs, of senior, customer-facing data scientists, with Big-Five traits and a high/low success outcome).

Several groups make expensive decisions from exactly this kind of information, usually by instinct: a learning-and-development head choosing which capability to fund, a student choosing what to learn, a candidate choosing which job to accept, and a university deciding what to teach. A wrong choice costs a training budget, a year of study, a resignation or a curriculum cycle.

1.2 Problem statement

PROBLEM STATEMENT

Which data-science capability should be developed next — and how confident can we be — when internal performance evidence, external market evidence and remaining headroom may point in different directions?

Table 2. Who decides, what it costs to be wrong.

Stakeholder	Decision they face	Cost of a wrong decision	What NEXUS DELTA gives them
L&D head / analytics leader	Which capability to fund for juniors	Budget spent on skills that do not move outcomes	PRIORITIZE / INVESTIGATE / MAINTAIN with evidence and confidence
Student / early professional	What to learn next for a target role	A year learning what the market discounts	Ranked development priorities for that role + salary-band distribution
Candidate / employee	Whether a role will fit how they work	Early attrition, friction, regret	Work-style fit, questions to ask, employee-reported culture
University placement cell	Which skills to add to the curriculum	Graduates trained for yesterday's roles	Capability demand by role and city tier

1.3 The workforce hierarchy

We organise the evidence as a hierarchy, borrowing the logic of Maslow's hierarchy of needs: a professional must first be employable, then grow, and only then can they choose where to thrive. Each level is answered by a different line of evidence, so the analysis never mixes them.

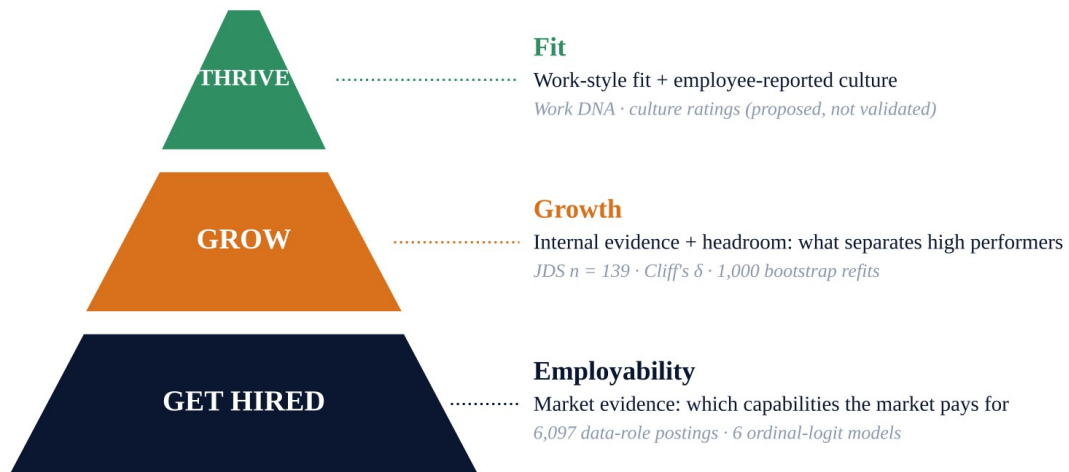


Figure 1. The NEXUS DELTA workforce hierarchy — a framing device, not a finding. Each level is answered by its own evidence line.

1.4 Analytics objectives

Table 3. Analytics objectives and success criteria.

#	Objective	Measurable success criterion	Answered in
O1	Measure which capabilities separate high- and low-hike juniors	Effect size with 95% CI and multiple-testing correction	§4.1
O2	Measure which capabilities the market pays for	Odds ratios stable across ≥ 6 model specifications	§4.4
O3	Quantify remaining development headroom per capability	Share of juniors below ceiling + modelled gain with CI	§4.7
O4	Turn three signals into one auditable decision	Printed rule; winner must beat runner-up in $\geq 80\%$ of bootstraps	§4.7 , §4.9
O5	Help a student read any job description	Salary-band model beating baselines on grouped CV	§4.6
O6	Show where Bharat's opportunity gap is	Capability and pay differences by city tier	§3.9

1.5 Hypotheses

- **H1** — A capability that separates high performers internally is also paid a premium by the market.
- **H2** — Big Data, widely promoted as a core data-science skill, separates high performers.
- **H3** — The text of a job posting (title and listed skills) predicts its salary band beyond experience alone.
- **H4** — Seniority carries a consistent pay premium across data roles.
- **H5** — Personality traits separate successful senior, customer-facing data scientists.

AN OPEN QUESTION, ANSWERED IN §4.5

If H1 is true, every strong internal skill should be a market premium. **One of them is not.** Which one — and why — turned out to be the most useful finding in this study ([§4.5](#)).

1.6 Scope

In scope: the five technical capabilities measured in JDS (Big Data, Maths & Statistics, Coding, AI & ML, Dashboards & Storytelling), the Indian analytics job market as described in the brief (postings from 2024–25), and a decision rule an organisation can audit. Out of scope: causal claims (all evidence is associational), individual salary prediction from JDS (one company, $n = 139$), and any hiring or rejection decision about a person.

2. Approach Description

IN ONE LINE

Three signals, kept apart and combined by written rules. Agreement → PRIORITIZE.
Disagreement → INVESTIGATE. A weak internal link → MAINTAIN.

2.1 The Zero Rule: why we do not join the datasets

The obvious first step — merging the four files — is invalid. Our identifier audit (§3.5) found **0 IDs shared between JDS and SDS**; the overlaps that do exist (for example 1,460 between DS Jobs reference numbers and Analytics Jobs serial numbers, and 137 and 152 between Analytics serial numbers and JDS/SDS IDs) are coincidental number ranges, not shared entities. The files also describe different grains: a posting, a company opening, a junior employee, a senior employee. Joining them would fabricate relationships.

Instead, NEXUS DELTA connects the datasets through a **versioned capability taxonomy (cap-tax-v2.0)**: the same five capabilities are measured as scores in JDS and detected as keyword (regex) flags in posting text. Each dataset answers the question it can actually answer.

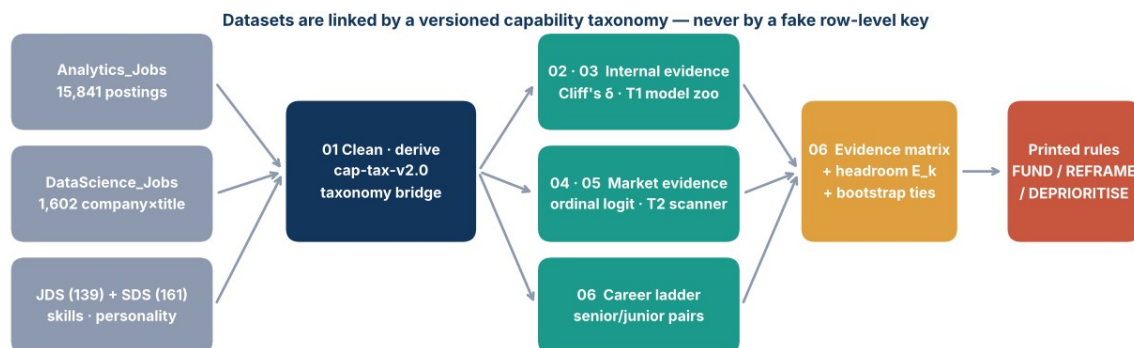


Figure 2. The NEXUS DELTA pipeline: data → evidence → insight → decision → action. Datasets meet only at the capability level. Notebook verdicts FUND / FUND WITH REFRAME / DEPRIORITISE appear in the product as PRIORITIZE / INVESTIGATE / MAINTAIN.

2.2 Overall flow

Table 4. Pipeline stages. Notebook links open the executed code.

Stage	Notebook	Input	Output	Why this stage exists
1 Explore & clean	NB01	4 raw files	analytics_clean.csv, taxonomy.json	Fix duplicates, parse fields, flag bad rows, prove no join
2 Internal evidence	NB02	JDS, SDS	Cliff's δ + CI + Holm	What separates high performers inside one company
3 Predictive check	NB03	JDS	7-model zoo, permutation test	Is the internal signal real or noise?
4 Market evidence	NB04	Analytics Jobs	Ordinal-logit odds ratios × 6 specs	What the market pays for, with controls

Stage	Notebook	Input	Output	Why this stage exists
5 JD scanner	NB05	Analytics Jobs	T2 model zoo, selected model	Read any job description for a student
6 Headroom & decision	NB06	Outputs of stages 2–5, DS Jobs	E_k what-if, bootstrap ranking, ladder, SQLite database	Combine signals into one auditable decision
7 Role profiles	NB07	Analytics Jobs	Role demand, band shares	Feed the student lens of the app
8 Supplementary checks	NB08	JDS, SDS, Analytics Jobs	Ranking detail, cut-point check, tier ratios, descriptives	Robustness and traceability for every number quoted

2.3 Why three signals instead of one score

A weighted average would hide the most important information: disagreement. Storytelling scores HIGH internally and LOW in the market; averaged, it looks medium — and an organisation could fund report production that the market discounts. NEXUS DELTA therefore keeps the signals separate and makes the conflict visible ([see the Evidence Convergence figure, §4.9](#)).

Table 5. The three signals and how each is graded.

Signal	Source	Statistic	HIGH when
Internal evidence	JDS (n = 139)	Cliff's δ , Mann–Whitney U, 2,000-bootstrap CI, Holm	$\delta \geq 0.33$ and 95% CI above 0 (LOW: Holm $p \geq 0.05$ or $\delta < 0.147$)
Market evidence	Analytics Jobs: 6,097 data-role postings (10,338 in the wide check)	Proportional-odds ratio for a higher salary band	Premium (CI > 1) in both primary (S5) and wide (S6) models; LOW = discount in both
Headroom	JDS	% of juniors below 5.0; E_k = modelled gain per +0.5 step	$\geq 60\%$ (MEDIUM 45–60%, LOW < 45%); E_k ranks funded capabilities

2.4 Decision rules

Table 6. The decision rule (NB06). Thresholds follow Romano et al. for Cliff's δ (0.147 small, 0.33 medium, 0.474 large); ‘strong’ therefore means at least a medium effect with a confidence interval above zero.

Condition	Verdict	Product label
Strong internal + market premium or neutral	FUND	PRIORITIZE
Strong internal + market discount in both taxonomies	FUND WITH REFRAME	INVESTIGATE
Moderate internal + robust premium	MONITOR	INVESTIGATE
Weak internal (Holm $p \geq 0.05$ or $\delta < 0.147$), or moderate without a premium	DEPRIORITISE	MAINTAIN
Among funded: winner must beat runner-up in $\geq 80\%$ of 1,000 paired bootstraps	else TIED	INVESTIGATE (human choice logged)

2.5 Method choices and the reason for each

Table 7. Methods and justification.

Method	Used for	Why this, not the alternative
Mann–Whitney U + Cliff's δ	JDS/SDS group differences	Scores are bounded 1–5 and ceiling-heavy; δ is distribution-free and interpretable, t-tests assume normality
Holm correction	5 simultaneous tests per family (JDS, SDS)	Controls family-wise error and is uniformly more powerful than Bonferroni
L2 logistic + StratifiedGroupKFold	T1 high-hike model	Duplicate employee profiles exist; grouping prevents the same profile in train and test
Permutation test (300 shuffles for T1, 40 for T2)	T1 and T2 validity	Shows the scores are not achievable by chance
Ordinal logit (statsmodels OrderedModel)	Market evidence	Salary bands are ordered; ordinal logit uses the order, multinomial throws it away
6 nested specifications	Robustness	A skill only counts as a premium if it survives experience, seniority, location, role and data-quality controls
TF-IDF (title + listed skills) + structured logistic	T2 JD scanner	Best of 7 models on grouped CV; linear, fast, explainable, runs offline
Paired bootstrap ranking	Final decision	Turns ‘which is first?’ into a probability; declares ties instead of hiding them

Key formulas

Cliff's δ (internal evidence), for high-hike scores x_i and low-hike scores y_j :

$$\delta = [\#(x_i > y_j) - \#(x_i < y_j)] / (n_{\text{high}} \cdot n_{\text{low}}) = 2U / (n_{\text{high}} \cdot n_{\text{low}}) - 1, \quad -1 \leq \delta \leq 1$$

Proportional-odds ordinal logit (market evidence), for salary band $Y \in \{0, \dots, 5\}$, capability flags S_k and controls Z :

$$\text{logit } P(Y \geq j) = \alpha_j + \sum_k \beta_k S_k + \gamma Z, \quad OR_k = \exp(\beta_k) \quad (OR_k > 1: \text{higher bands more likely})$$

Modelled gain E_k (headroom), using the fitted T1 logistic model $p(\cdot)$ on all 139 juniors (juniors already at 5.0 contribute zero):

$$E_k = 100 \cdot \text{mean over juniors} [p(x \text{ with } x_k \rightarrow \min(x_k + 0.5, 5)) - p(x)] \quad (\text{percentage points})$$

Decision confidence, from $B = 1,000$ paired bootstrap refits:

$$P(\text{beat}) = (1/B) \cdot \sum_b 1 [E_{\text{lead}}(b) > E_{\text{next}}(b)], \quad \text{TIED if } P(\text{beat}) < 0.80$$

2.6 Validation strategy

- **Leakage control:** grouped cross-validation on duplicate profiles for T1; normalised job-title groups for T2 (6,741 groups across 10,338 postings).
- **Chance control:** permutation tests for T1 and T2; Holm correction for multiple tests.
- **Robustness:** six market specifications, two taxonomies, a subset without lead/manager/head titles, a subset without promotional titles, and a proportional-odds check by cut-point.
- **Calibration:** predicted vs observed probabilities for T1 and T2.
- **Software:** 171 automated checks on the app, including edge cases (missing model, corrupt results, empty input).

2.7 Technology

Python (pandas, scipy, statsmodels, scikit-learn) in eight Colab-ready notebooks with a fixed random seed (42); SQL queries on a SQLite database of the cleaned tables and the evidence

matrix; a SAS mirror of the core steps for SAS VFL (a template, not executed); and a Streamlit application that runs fully offline. Everything is in the [public repository](#).

3. Data Exploration

IN ONE LINE

87% of skill fields were truncated, 1,001 postings were duplicates, and 25 junior profiles contradicted each other. Every problem was measured and treated before modelling.

3.1 Inventory

Table 8. Datasets supplied in the brief.

File	Rows	Grain	Key fields	Role in the study
Analytics_Jobs.csv	15,841	One posting	designation, key_skills, description, experience, salary band, location	Market evidence; JD scanner
DataScience_Jobs.csv	1,602	Company × title opening	company, title, min experience, avg/min/max salary, num_of_jobs	Career ladder; salary context
JDS_Skill_Traits.xlsx	139	Junior employee	5 skill scores (1–5), high/low hike	Internal evidence; headroom
SDS_Personality_Traits.xlsx	161	Senior employee	Big-Five traits (normalised), high/low success	Personality context

3.2 Data quality audit

Table 9. Data-quality issues, their size and the treatment. ‘Raw’ = 15,841 supplied rows; ‘clean’ = 14,840 after de-duplication.

Dataset	Issue found	Evidence	Impact if ignored	Treatment
Analytics Jobs	Exact duplicate postings	1,001 raw rows (6.3%)	Inflated counts, leakage in CV	Dropped; job-title groups in CV
Analytics Jobs	Skills field truncated (‘...’)	13,806 raw rows (87.2%); 87.0% after de-duplication	Missed skills → biased toward null	Truncation flag as control; description text parsed in the wide taxonomy (S6)
Analytics Jobs	Job type missing	11,567 clean rows (77.9%)	Unusable as feature	Missingness flag as control only
Analytics Jobs	Description missing	3,508 raw rows (3,371 clean)	Fewer text cues	Kept; title and skills fields used
Analytics Jobs	Promotional titles (‘urgent’, ‘walk-in’, ₹)	1,722 clean rows (586 in data roles)	Noise in role mix	Flagged; robustness run without them (max ΔOR 0.066)
Analytics Jobs	Salary given only as 6 bands	all rows	No exact salary exists	Ordinal models; distributions, never exact pay
Analytics Jobs	Free-text locations	all rows	City inconsistency	Tier-1 / Tier-2-3 / International by regex
JDS	Duplicate skill profiles	29 groups, 62 rows	Same profile in train and test	StratifiedGroupKFold; de-duplicated re-check
JDS	Identical profiles, opposite outcomes	12 groups, 25 rows	Ceiling on accuracy	Reported; caps achievable AUC
JDS	Ceiling at 5.0	58% of juniors at 5.0 in Storytelling	Little room left to grow	Measured as ‘headroom’

Dataset	Issue found	Evidence	Impact if ignored	Treatment
SDS	Repeated IDs	161 rows, 152 unique IDs	Double counting	Grouped CV for the model; rank tests use all rows (9 repeats, 5.6%)
All	No shared identifiers	$JDS \cap SDS = 0$	Invalid joins	Taxonomy bridge (the Zero Rule)

3.3 Dataset profiles, outliers and inconsistencies

Table 10. Profiling every file: outliers, inconsistencies and what we did.

Dataset	Check	Finding	Decision
Analytics Jobs	Category inconsistency	job_type has 5 spellings of one value (Analytics 2,971 · analytics 746 · ANALYTICS 64 · analytic 30 · Analytic 19); 12,011 raw rows blank	Normalised to lower-case ‘analytics’; missingness kept as a flag
Analytics Jobs	Location text	1,355 distinct location strings; 2,553 postings list several cities; 313 distinct spellings contain Bangalore/Bengaluru	Regex tiering on the full string (Tier-1 if any Tier-1 city appears)
Analytics Jobs	Experience range	Ranges up to 30 years; spans up to 10 years; 0 unparseable	Kept; span used as a control
Analytics Jobs	Salary distribution	6 bands; after de-duplication the mode is 10–15 LPA (3,503) and 25–50 LPA holds 992 (1,245 raw — a fifth of top-band rows were duplicates)	Ordinal target; no imputation of exact pay
DS Jobs	Text-typed numbers	Salaries stored as text (‘7.8L’)	Parsed to numeric LPA
DS Jobs	Outliers	39 average salaries above the IQR fence (Q1 7.6, Q3 17.2, max 82 LPA); 10 rows with > 1,000 openings	Kept — real senior roles; ladder uses medians and paired ratios, which are robust
DS Jobs	Consistency	$\min \leq \text{avg} \leq \max$ holds in all 1,602 rows; 134 reference numbers reused across 276 rows; 10 titles × 642 companies	Company × title pairs used for the ladder
JDS	Column names	‘maths-stats_skills’ uses a hyphen; outcome stored as 0/1	Renamed to snake_case keys
JDS	Range & ceiling	All scores within 1–5; 18–58% of juniors already at 5.0 per capability; 2 repeated IDs	Ceiling measured as headroom; grouped CV
SDS	Column names	Leading space in ‘extraversion’; ‘success_classification_high_low’ has embedded spaces	Stripped and renamed
SDS	Repeated IDs	9 repeated IDs (161 rows, 152 unique)	Grouped analysis

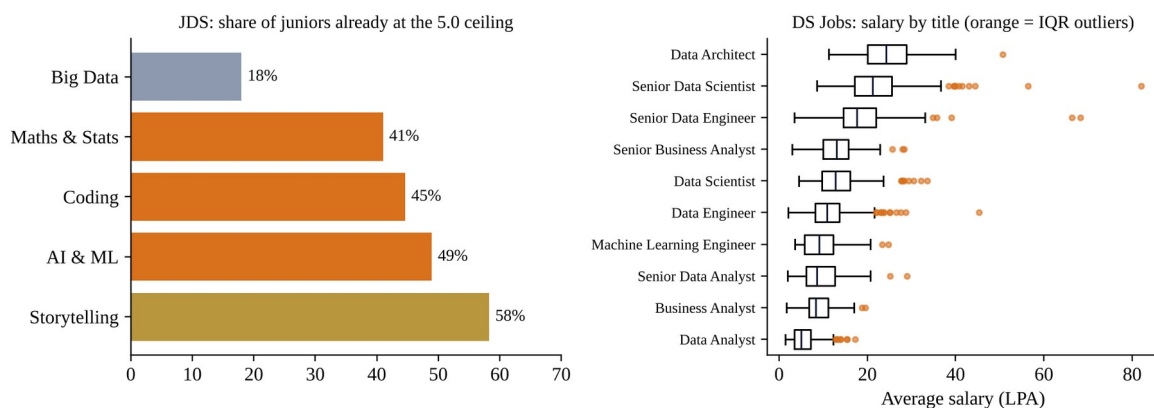


Figure 3. Two exploration findings that shaped the analysis: the JDS ceiling effect (left) and salary outliers by title in DS Jobs (right).

3.4 Descriptive statistics

Table 11. JDS skill scores, n = 139. High-hike share 52.5% (balanced outcome).

JDS capability (1–5)	Mean	SD	Min	Median	Max	Skew	% at 5.0
Big Data	3.85	0.85	2.3	3.8	5.0	−0.08	18.0%
Maths & Statistics	4.29	0.84	2.2	4.6	5.0	−0.97	41.0%
Coding	4.27	0.89	2.2	4.6	5.0	−0.89	44.6%
AI & ML	4.57	0.67	2.2	4.9	5.0	−1.70	48.9%
Dashboards & Storytelling	4.36	0.93	2.3	5.0	5.0	−1.08	58.3%

Four of the five skills are strongly left-skewed with medians at or near the 5.0 ceiling; only Big Data is symmetric. This has three consequences for the analysis: (1) mean-based tests would be distorted, so we use rank-based Mann–Whitney U and Cliff's δ ; (2) the 18 AI & ML scores flagged by the IQR rule are low scorers in a ceiling-compressed distribution, not errors, so they are kept; (3) the share already at 5.0 is itself a finding — it limits how much a capability can still improve, which we measure as headroom.

Table 12. SDS trait scores, n = 161. Success share 52.8%. Five agreeableness values fall outside the IQR fence and are kept (plausible range).

SDS trait (normalised)	Mean	SD	Min	Median	Max
Neuroticism	36.2	11.3	17	34	68
Extraversion	43.2	12.1	17	45	67
Openness	41.3	11.3	18	44	65
Agreeableness	44.6	11.3	17	46	68
Conscientiousness	45.2	13.2	18	49	66

Correlation between skills

Table 13. Rank correlations between JDS skills. Variance inflation factors are 1.04–1.45.

Spearman ρ	Big Data	Maths & Stats	Coding	AI & ML	Storytelling
Big Data	1.00	−0.18	0.05	−0.17	0.05
Maths & Statistics	−0.18	1.00	0.40	0.44	0.36
Coding	0.05	0.40	1.00	0.16	0.47
AI & ML	−0.17	0.44	0.16	1.00	0.27
Dashboards & Storytelling	0.05	0.36	0.47	0.27	1.00

The skills are moderately correlated at most (largest $\rho = 0.47$, Coding with Storytelling) and every variance inflation factor is below 1.5, so the multivariable T1 model can separate their contributions without collinearity problems. Big Data is nearly uncorrelated with everything else — consistent with it behaving as a separate, weakly related skill rather than part of the analytical core.

3.5 Identifier audit

Table 14. Identifier overlap across files (NB01).

ID comparison	Overlap	Interpretation
DS.reference_no $\cap$ A.s_no	1,460	coincidental row numbers — different grains

ID comparison	Overlap	Interpretation
DS.reference_no $\cap$ JDS.id	21	coincidental row numbers — different grains
DS.reference_no $\cap$ SDS.id	33	coincidental row numbers — different grains
A.s_no $\cap$ JDS.id	137	coincidental row numbers — different grains
A.s_no $\cap$ SDS.id	152	coincidental row numbers — different grains
JDS.id $\cap$ SDS.id	0	no shared entities

3.6 Derived variables

Table 15. Derivations (NB01, NB06).

Derived field	Rule	Example	Used in
salary_ord	6 ordered bands: 0–3, 3–6, 6–10, 10–15, 15–25, 25–50 LPA	‘10to15’ $\rightarrow$ 3	Market model, JD scanner
exp_min / exp_max / exp_span	Regex on ‘x–y yrs’ (0 parse failures)	‘5-10 yrs’ $\rightarrow$ 5, 10, 5	Controls, scanner
seniority	First match: head > lead/manager > senior > junior > mid	‘Sr Data Scientist’ $\rightarrow$ senior	Controls, ladder
role_family	DS/ML, DE, BA, DA/BI, other	‘Big Data Engineer’ $\rightarrow$ DE	Controls, role profiles
loc_tier	Tier-1 city list / foreign list / else Tier-2-3	‘Indore’ $\rightarrow$ Tier2_3	Bharat analysis
Capability flags $\times$ 7	cap-tax-v2.0 regex: primary = title + listed skills; wide = also the description	‘python, scikit-learn’ $\rightarrow$ coding, ai_ml	Market model
trunc_skills, jobtype_missing, spam_title	Data-quality flags	—	Controls, robustness
headroom_k	Share of juniors with score < 5.0 for capability k	Big Data 82%	Decision

3.7 Consolidation and data reduction

Table 16. Cleaning log, Analytics Jobs.

Step	Rows
raw rows	15,841
after exact-duplicate drop	14,840
experience parse failures	0
after dropping unparseable experience	14,840
Primary data-role population (is_ds_p)	6,097
Wide population (is_ds_w)	10,338

Data reduction. Rows: 15,841 $\rightarrow$ 14,840 after duplicates $\rightarrow$ 6,097 primary data roles for the main market model, with the 10,338-posting wide population kept as a robustness check. Features: free-text skills and descriptions are reduced to 7 capability flags; free-text locations to 3 tiers; designations to 5 seniority levels and 5 role families; salary to 6 ordered bands. Each reduction is a published rule in taxonomy.json, so it can be audited and re-run.

The cleaned postings, JDS, DS Jobs and the capability evidence matrix are written to one SQLite database (out/nexus_delta.db) and queried with standard SQL; the same tables can be loaded into SAS VFL. The capability prevalence in primary data roles is: Coding 29.4%, Maths

& Statistics 17.2%, Storytelling 16.5%, AI & ML 14.4%, MIS 13.0%, Big Data 9.5%, Visualisation 5.4%.

Excerpt — NB01 cleaning

```
a = A_RAW.drop_duplicates(subset=key_cols, keep="first").copy() # -1,001 rows
a["trunc_skills"] = a.key_skills.fillna("").str.rstrip().str.endswith("...") # ~87%
m = a.experience.str.extract(r"(\d+)\s*-\s*(\d+)").astype(float) # '5-10 yrs' -> 5, 10
a["salary_ord"] = a.salary.map({b: i for i, b in enumerate(BANDS)}).astype(int)
a["loc_tier"] = np.where(loc.str.contains(T1), "Tier1",
                        np.where(loc.str.contains(FOREIGN), "International", "Tier2_3"))
```

3.8 What exploration showed

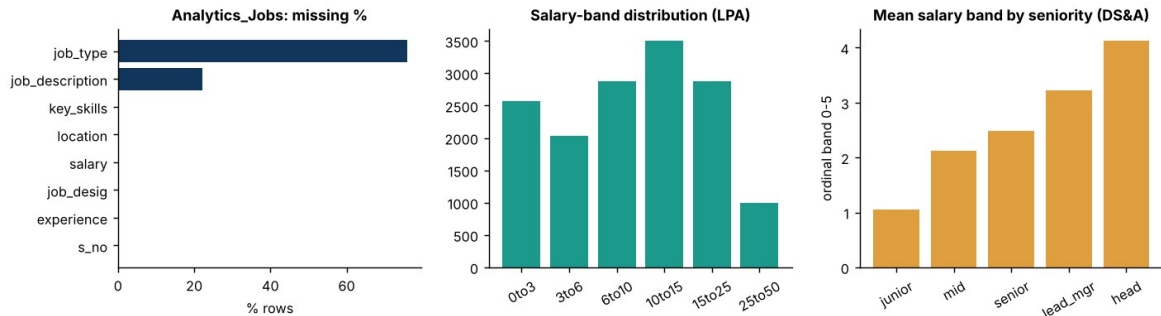


Figure 4. Exploration overview (NB01): missing-value rate by column, salary-band distribution, and mean salary band by seniority.

3.9 Bharat lens: the tier gap

Table 17. Opportunity by city tier, primary data-role postings (n = 6,097).

City tier	Postings	% at ≥ 10 LPA	AI & ML %	Coding %	Maths %
Tier1	5,627	51.0%	15.0	30.6	17.8
Tier2_3	407	29.0%	6.9	16.0	9.8
International	63	79.4%	4.8	7.9	9.5

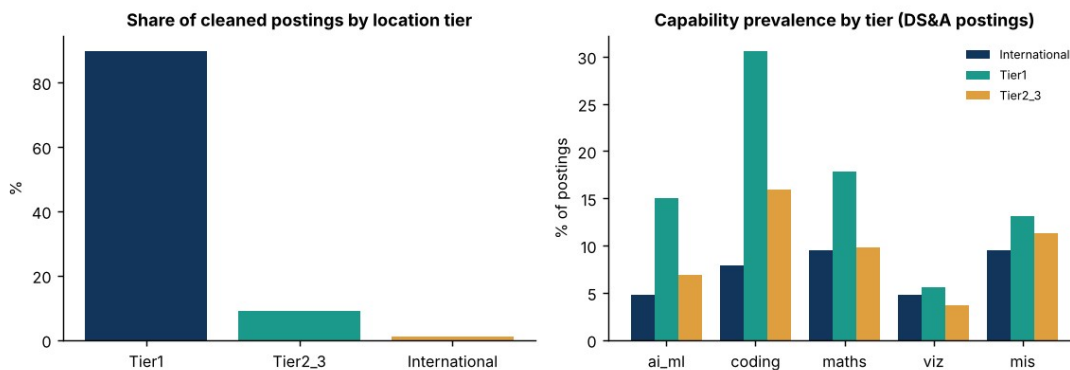


Figure 5. Share of postings by city tier (left) and capability prevalence by tier (right). In Tier-2/3 cities, 29% of data-role postings reach 10 LPA, about half the Tier-1 rate.

Table 18. Capability prevalence by tier, primary data-role postings (NB08).

Capability keywords	Tier-1 %	Tier-2/3 %	Tier-2/3 ÷ Tier-1
Big Data	9.8	6.1	0.63×
Maths & Statistics	17.8	9.8	0.55×
Coding	30.6	16.0	0.52×
AI & ML	15.0	6.9	0.46×

Capability keywords	Tier-1 %	Tier-2/3 %	Tier-2/3 ÷ Tier-1
Storytelling (family)	16.7	14.7	0.88×
Visualisation	5.6	3.7	0.66×
MIS / reporting	13.1	11.3	0.86×
MIS only	11.2	11.1	0.99×

The skill mix differs by tier. Tier-2/3 postings mention Maths & Statistics, Coding and AI & ML only about half as often as Tier-1 postings (0.55×, 0.52×, 0.46×), while MIS/reporting keywords are almost as common (0.86×; MIS-only 0.99×). Tier-2/3 data roles therefore lean toward the reporting work that the market prices lower (\$4.5) — part of why only 29% of them reach 10 LPA.

4. Data Analysis

IN ONE LINE

Maths & Statistics clears all three tests and has the largest modelled gain. Storytelling is strongest inside the company but discounted outside — and the analysis shows where the discount comes from.

4.1 Internal evidence: what separates high-hike juniors

Table 19. JDS, n = 139, 52.5% high hike. Mann–Whitney U, 2,000-bootstrap CI, Holm-corrected.

Capability	Mean (low hike)	Mean (high hike)	Cliff's δ	95% CI	Holm p
Dashboards & Storytelling	3.81	4.85	0.59	[0.44, 0.72]	1.2e-10
Maths & Statistics	3.83	4.71	0.55	[0.40, 0.70]	2.7e-08
Coding	3.85	4.64	0.49	[0.31, 0.64]	6.1e-07
AI & ML	4.28	4.82	0.36	[0.19, 0.52]	1.7e-04
Big Data	3.75	3.94	0.12	[-0.09, 0.32]	0.22

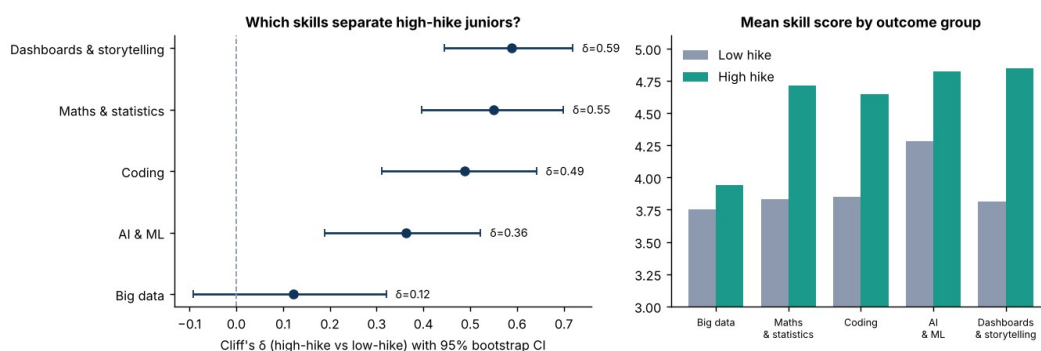


Figure 6. Internal effect sizes with 95% bootstrap intervals. Big Data's interval crosses zero.

Four of five capabilities separate high- and low-hike juniors with medium-to-large effects. **H2 is not supported:** Big Data does not separate them on its own ($\delta = 0.12$, CI -0.09 to 0.32 , Holm $p = 0.22$). Note that in the multivariable T1 model (§4.2) Big Data does carry a positive coefficient once the other four skills are held fixed: it is slightly negatively correlated with Maths & Statistics and AI & ML ($\rho \approx -0.18$), which masks its effect in the univariate test (a suppression effect). Its univariate null result should be revisited with more data.

4.2 Is the internal signal real? (T1 model zoo)

Table 20. T1: predicting high hike from the five skills (NB03).

Model	AUC (grouped CV)	Brier	Accuracy
Prevalence baseline	0.491 ± 0.000	0.249	0.525
Storytelling-only LR	0.772 ± 0.008	0.169	0.763
kNN (k=15)	0.875 ± 0.006	0.147	0.800
Random forest	0.893 ± 0.006	0.127	0.847
Gradient boosting	0.880 ± 0.009	0.130	0.829
RBF-SVM	0.886 ± 0.011	0.119	0.853
NEXUS L2 logistic (C=1)	0.902 ± 0.005	0.116	0.863

The regularised logistic model (AUC 0.90) beats random forest, boosting, SVM and kNN, and is the most interpretable. After removing duplicate profiles the AUC is still **0.88** ($n = 118$). A permutation test gives $p \leq 0.0033$ (the smallest attainable with 300 shuffles): the best shuffled run reached only 0.66. The learning curve rises from 0.876 at $n = 40$ to 0.901 at $n = 100$ and only 0.904 at $n = 139$, so more data from the same company would add little.

Table 21. What drives the T1 model.

Capability	Coefficient (std.)	Permutation importance (AUC drop)
Maths & Statistics	1.28	0.110
Dashboards & Storytelling	1.12	0.095
AI & ML	0.76	0.037
Big Data	0.68	0.028
Coding	0.53	0.015

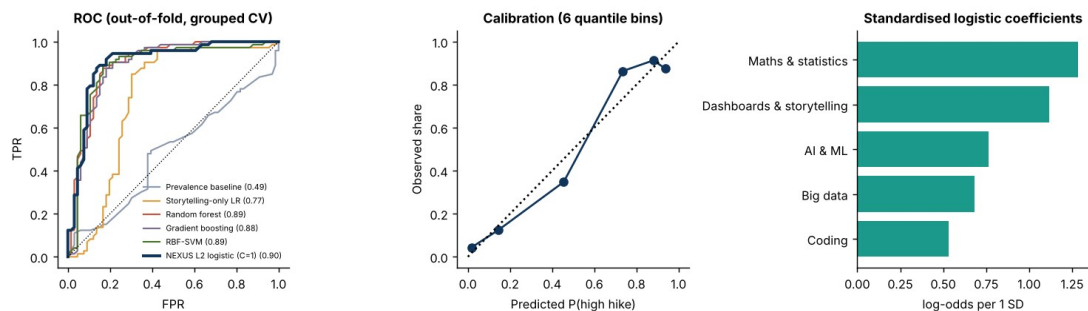


Figure 7. T1 model comparison, calibration and drivers.

4.3 Personality context (SDS)

Table 22. SDS, $n = 161$ rows (152 unique IDs). Cross-validated AUC 0.96.

Trait	Mean (low success)	Mean (high success)	Cliff's δ	Holm p
Openness	33.3	48.5	0.77	<0.001
Conscientiousness	35.7	53.7	0.76	<0.001
Extraversion	36.9	48.9	0.57	<0.001
Agreeableness	41.1	47.7	0.30	0.002
Neuroticism	36.3	36.1	0.07	0.454

H5 is supported: openness (δ 0.77), conscientiousness (0.76) and extraversion (0.57) separate successful senior, customer-facing data scientists; neuroticism does not. Rank tests use all 161 rows; the 9 repeated IDs (5.6%) are handled by grouping in the cross-validated model. We use this only as context. SDS has no link to JDS (the Zero Rule), and normalised trait scores are not an emotional-intelligence measure, so NEXUS DELTA never scores individuals on personality.

4.4 Market evidence: what the market pays for

Table 23. Proportional-odds ratio for a higher salary band, six ordinal-logit specifications on 6,097 data-role postings (S6: 10,338) (NB04). S1 flags only · S2 +experience · S3 +seniority · S4 +location · S5 all controls · S6 wide taxonomy & population.

Capability	S1	S2	S3	S4	S5	S6
Maths & Statistics	1.60	1.41	1.31	1.30	1.27	1.42
AI & ML	1.93	2.08	2.13	2.13	1.38	1.25
Coding	1.54	1.52	1.53	1.53	1.52	1.55
Dashboards & Storytelling	0.69	0.75	0.74	0.74	0.75	0.80
Big Data	1.77	1.16	1.16	1.16	1.09	1.15

Maths & Statistics, Coding and AI & ML are significant premiums in **6 of 6** specifications. AI & ML's premium falls from 2.13 to 1.38 once role family and data-quality controls are added (S5) — part of it is carried by job titles. Big Data's premium (1.77 unadjusted) attenuates to 1.16 and loses significance (CI 0.99–1.37) once experience is added, consistent with an experience proxy rather than a skill premium. Excluding lead/manager/head titles ($n = 4,816$) gives the same picture (Maths 1.37, Coding 1.62, AI & ML 1.61, Storytelling 0.74), and dropping 586 promotional titles changes no odds ratio by more than 0.066.

Table 24. Proportional-odds check: odds ratios from separate binary logits at each band cut-point, S5 controls (NB08). Bold = 95% CI excludes 1.

Capability	≥ 3 LPA	≥ 6 LPA	≥ 10 LPA	≥ 15 LPA	≥ 25 LPA
Maths & Statistics	1.80	1.77	1.21	1.17	0.97
AI & ML	22.83	1.57	1.15	1.52	1.69
Coding	2.44	1.76	1.43	1.28	1.15
Dashboards & Storytelling	0.76	0.66	0.74	0.77	0.83
Big Data	2.55	1.67	1.04	0.86	1.07

Proportional-odds check. The single odds ratios in §4.4 assume one effect across the whole pay range, so we re-estimated each capability at every cut-point. Storytelling stays below 1 at every cut (significant at the first four) and Coding stays above 1 at every cut (significant up to 15 LPA). Maths & Statistics is a significant premium through the 10 LPA cut, weaker at 15 LPA (CI crosses 1) and neutral at 25 LPA. AI & ML is above 1 at every cut but not significant at the 10 LPA cut; its ≥ 3 LPA estimate (22.8) rests on only 10 postings in the lowest band and is unstable. Big Data is significant only at the 6 LPA cut. The pooled odds ratios are therefore averages across the pay range, and we read them that way.

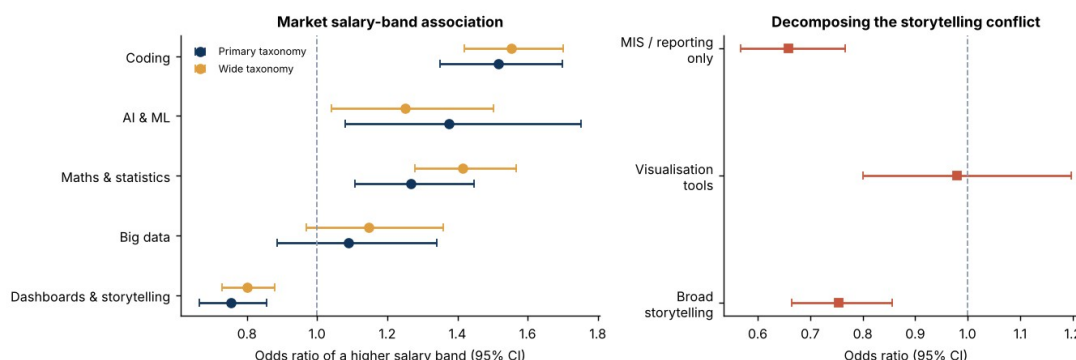


Figure 8. Market odds ratios with 95% CIs in the primary (S5) and wide (S6) models (left), and the Storytelling decomposition (right).

4.5 The Storytelling paradox — resolved

Storytelling is the strongest internal separator (δ 0.59) and a market **discount** in every specification (OR 0.69–0.80). **H1 does not hold for Storytelling.** As an exploratory follow-up, we split the storytelling keyword family into its two sub-taxonomies and re-fitted the S5 model on the 6,097 data-role postings:

Table 25. Decomposing the Storytelling discount.

Sub-family	Example keywords	Odds ratio	95% CI	p
Visualisation	Tableau, Power BI, dashboards, storytelling	0.98	[0.80, 1.20]	0.84
MIS / reporting only	MIS, Excel reporting, report generation	0.66	[0.57, 0.77]	5.7e-08

THE ANSWER TO THE OPEN QUESTION

The discount is not attached to communication as such. Postings with visualisation keywords are priced neutrally (OR 0.98, $p = 0.84$); postings with MIS/reporting-only keywords are associated with lower bands (OR 0.66). So the decision is neither ‘fund’ nor ‘cut’:

INVESTIGATE — fund with reframe: develop analytical storytelling, not report production. An averaged score would have hidden this.

This result depends on how the taxonomy defines storytelling. With visualisation keywords alone the market signal is neutral in the primary model (although its interval, 0.80–1.20, cannot rule out a modest discount); the discount comes mainly from the MIS/reporting keywords. That is why the verdict is to reframe the capability rather than to cut it.

4.6 Reading a job description: the T2 scanner

Table 26. T2: salary band from a job posting; grouped CV on 10,338 postings in 6,741 job-title groups (NB05).

Model	Exact band	Within 1 band	Quadratic κ	MAE (bands)
B0 majority	23.9%	63.9%	0.000	1.28
B1 experience-only LR	38.3%	81.1%	0.655	0.85
M0 structured LR (no flags)	41.4%	82.0%	0.677	0.82
M1 structured LR (+flags)	42.3%	82.6%	0.688	0.81
M2 TF-IDF + structured LR	44.4%	85.1%	0.719	0.75
M3 random forest	41.9%	81.5%	0.672	0.83
M4 hist-gradient boosting	42.0%	81.5%	0.672	0.83

H3 is supported. The text of the posting (TF-IDF on title and listed skills, plus structured fields) beats the experience-only model on every metric: quadratic κ 0.719 vs 0.655, exact band 44.4% vs 38.3%, within one band 85.1% vs 81.1%. Adding the capability flags to the structured fields alone raises κ by 0.011 (95% bootstrap CI 0.006–0.016), and a label-permutation test of that model never exceeded κ 0.008 (observed 0.688). Because 56% of exact-band predictions are still wrong, the product shows the full probability distribution and never an exact salary.

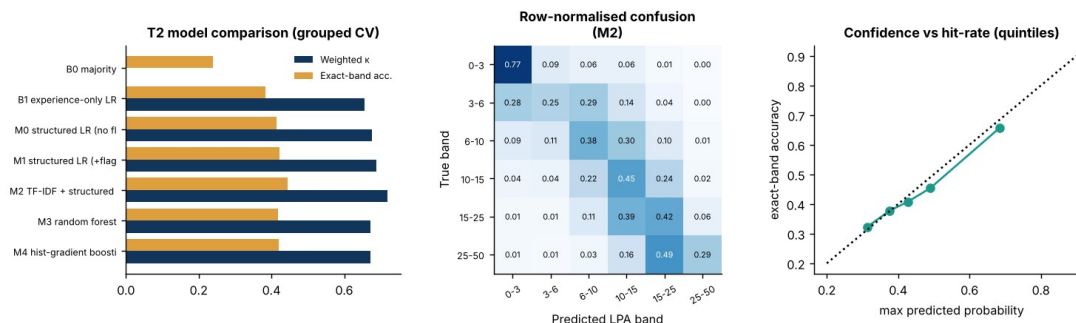


Figure 9. T2 model comparison, normalised confusion matrix of the selected model, and confidence versus hit rate (calibration).

4.7 Headroom and modelled gain

Table 27. E_k = modelled rise in $P(\text{high hike})$ if every junior improved by up to 0.5 points in capability k (capped at 5.0), averaged over all 139 juniors (what-if, not causal).

Capability	Headroom	E_k (pp per +0.5 step)	95% CI	P(top)
Big Data	82%	3.47	[1.04, 6.06]	14.5%
Maths & Statistics	59%	5.15	[2.96, 7.47]	79.5%
Coding	55%	1.98	[0.13, 3.92]	2.0%
AI & ML	51%	2.77	[0.78, 4.59]	2.2%
Dashboards & Storytelling	42%	2.75	[1.49, 4.11]	1.8%

Maths & Statistics has the largest modelled gain (+5.1 pp, CI 3.0–7.5). Across 1,000 bootstrap refits it beats the runner-up among funded capabilities (AI & ML) in **95.7%**, ranks first among the four funded capabilities in **91.3%**, and ranks first among all five in 79.5% (NB06, NB08). Big Data has the second-largest E_k (3.5 pp, CI 1.0–6.1) but is not eligible: its univariate internal signal is not significant, so the rule (§2.4) sets it to MAINTAIN. The ranking was re-run over 9 settings (step size 0.25/0.5/1.0 $\times$ regularisation $C = 0.1/1/10$): **Maths & Statistics is first in all 9**. E_k is computed in-sample on 139 juniors and is a what-if, not a causal effect.

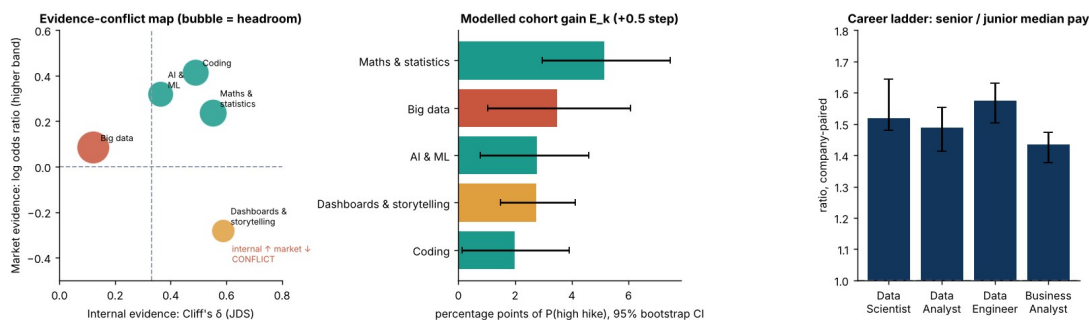


Figure 10. Evidence-conflict map (bubble = headroom), modelled gain E_k with 95% bootstrap CIs, and the career ladder (§4.8).

4.8 Career ladder and salary context

Table 28. Paired company-level comparison, Data Science Jobs (NB06).

Role family	Senior ÷ junior median salary	95% CI	Pairs	Senior higher	Wilcoxon p
Data Scientist	1.52×	[1.48, 1.64]	105	96%	4e-19
Data Analyst	1.49×	[1.41, 1.55]	111	96%	1e-19
Data Engineer	1.58×	[1.50, 1.63]	112	97%	3e-20
Business Analyst	1.43×	[1.38, 1.47]	124	95%	1e-21

H4 is supported: within the same company, senior roles pay about 1.4–1.6× junior roles, in 95–97% of pairs. Salary models on this file reach R^2 0.59 (gradient boosting, MAE 3.5 LPA), with title and experience explaining most of it — further reason not to promise individuals an exact number.

4.9 Evidence convergence: the decision

The three signals for each capability, side by side. Green is HIGH, gold MEDIUM, red LOW, graded as in §2.3 (Table 5). The arrow is the decision rule, not an average.

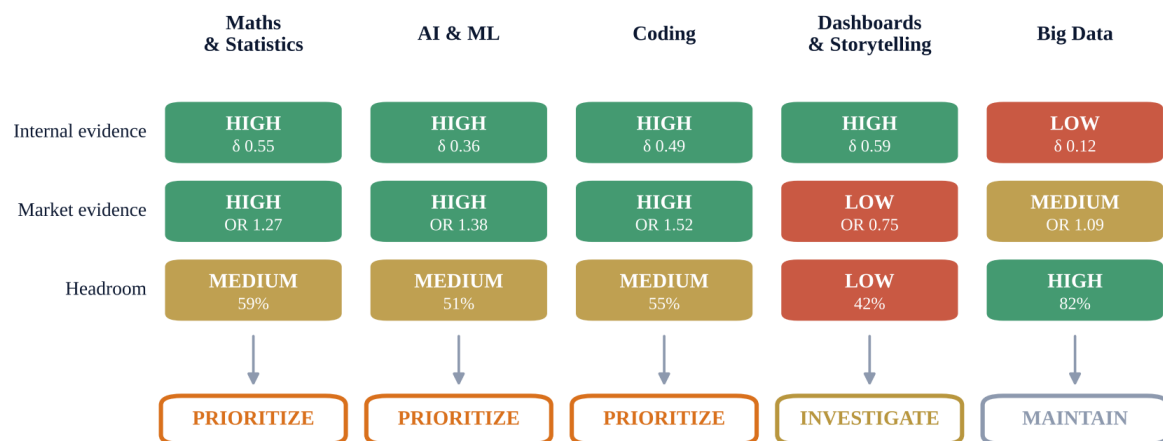


Figure 11. Evidence Convergence. Maths & Statistics, AI & ML and Coding converge; Storytelling conflicts; Big Data has headroom but weak evidence.

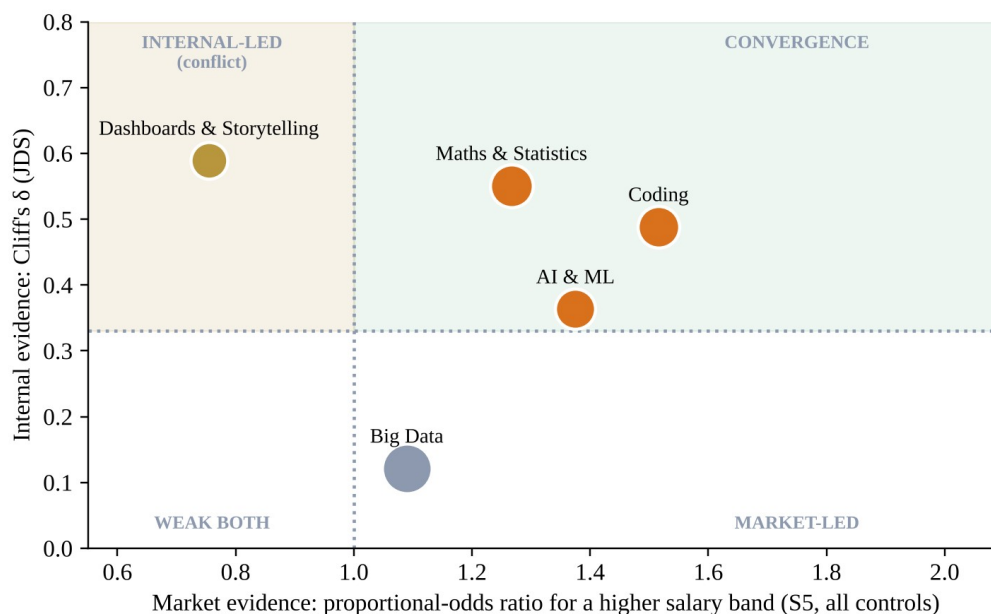


Figure 12. Evidence conflict matrix. Bubble size = headroom; colour = decision. Storytelling sits alone in the conflict quadrant. Big Data's odds ratio (1.09, CI 0.89–1.34) is not significant, so it is treated as neutral despite sitting right of 1.

5. Results & Conclusions

IN ONE LINE

Maths & Statistics is the best-supported next investment (first among funded capabilities in 91.3% of refits). Reframe Storytelling. Do not chase Big Data just because there is room.

5.1 The decision

Table 29. NEXUS DECISION.

#	Capability	Decision	Why
1	Maths & Statistics	PRIORITIZE	Strong internal (δ 0.55) + premium in 6/6 models + 59% headroom + largest modelled gain; beats AI & ML in 95.7% of refits
2	AI & ML	PRIORITIZE	Strong internal (δ 0.36, CI 0.19–0.52) + premium in 6/6 models; part of the premium is carried by job titles
3	Coding	PRIORITIZE	Strong internal (δ 0.49) + the largest controlled premium (OR 1.52); smaller modelled gain
4	Dashboards & Storytelling	INVESTIGATE	Strongest internal (δ 0.59) but market discount, concentrated in MIS/reporting keywords; reframe as analytical communication
5	Big Data	MAINTAIN	No significant univariate internal link (δ 0.12); market premium vanishes with experience controls; 82% headroom is not enough on its own

5.2 Objectives met

Table 30. Objectives and results.

#	Objective	Result
O1	Internal separation	4 of 5 capabilities separate high/low hike; Big Data does not (§4.1)
O2	Market value	3 premiums and 1 discount significant in all 6 specs; Big Data significant only unadjusted (1 of 6) (§4.4)
O3	Headroom	Measured per capability; E_k with bootstrap CI (§4.7)
O4	Auditable decision	Written rules; winner beats runner-up in 95.7% of refits (threshold 80%); first among funded in 91.3%; 9/9 settings agree
O5	Read any posting	44% exact / 85% within-one band; better than experience-only on all metrics
O6	Bharat gap	Tier-2/3: 6.7% of data-role postings; 29% vs 51% reach 10 LPA

5.3 Hypotheses

Table 31. Hypotheses and outcomes.

Hypothesis	Outcome	Evidence
H1 Internal strength = market premium	Partly supported	True for Maths, Coding, AI & ML; not for Storytelling (OR 0.75)
H2 Big Data separates performers	Not supported	δ 0.12, CI crosses 0, Holm p 0.22 (positive only conditionally, in T1)
H3 Posting text predicts salary band	Supported	κ 0.719 vs 0.655 (experience-only); exact 44.4% vs 38.3%
H4 Seniority premium is consistent	Supported	1.43–1.58 $\times$ in 95–97% of paired comparisons
H5 Personality separates senior success	Supported	Openness δ 0.77, conscientiousness 0.76; AUC 0.96

5.4 Conclusions

- Evidence beats intuition, and disagreement is information.** The single most useful finding — the Storytelling paradox — exists only because the signals were not averaged.
- Maths & Statistics is the best-supported investment** for junior data scientists: strong internally, a premium in every market model, the largest modelled gain, and first under every sensitivity setting we tried.
- Unadjusted signals mislead.** Big Data looks valuable in raw postings (OR 1.77) but the premium loses significance once experience is controlled, and it does not separate performers on its own.
- Students should be shown distributions, not promises.** A posting's title and skills predict its salary band usefully but imperfectly.
- Opportunity is unevenly distributed across Bharat:** Tier-2/3 data-role postings reach 10 LPA about half as often as Tier-1 postings, and lean toward MIS/reporting work rather than statistics, coding and AI & ML.

6. Implications

IN ONE LINE

The same evidence engine serves the organisation that funds skills, the student who learns them and the employee who lives with the job.

6.1 For each stakeholder

Table 32. Implications for each stakeholder.

Stakeholder	Implication	Action this week
L&D head	Make Maths & Statistics the first candidate for the next training budget, and measure the result; redesign Storytelling training around analytical narrative, not MIS reports	Run NEXUS DECISION on your own skills data; review any ties with the team
Student	Learn statistics first; build real AI & ML and coding skill rather than chasing titles (part of the AI & ML premium is carried by job titles); be cautious about MIS-only roles if pay matters	Scan three target job descriptions; follow the priority tiers
University placement cell	Add applied statistics and communication-of-analysis to the curriculum	Map courses to the five capabilities; track by role
Hiring manager	Write job descriptions that match the real work; the market already prices titles	Use Work DNA questions in interviews — as discussion, never as a filter
Employee	Rate how the work really is, anonymously, so the next candidate knows	Join with an invite code; nothing is shown until 5 responses exist
Bharat / policy	29% of Tier-2/3 data-role postings reach 10 LPA vs 51% in Tier-1	Expand applied statistics and coding programmes outside metros

6.2 From analysis to product

NEXUS DELTA is not a report that ends here. The analysis drives a working, offline Streamlit application ([source](#)) with three lenses: **Organisation** (Evidence Convergence and NEXUS DECISION), **Student** (career priorities and a JD scanner) and **Fit** (Work DNA and employee-reported culture — proposed instruments, not yet validated). Every number in the app carries a provenance badge — source file, notebook, method and limitation.

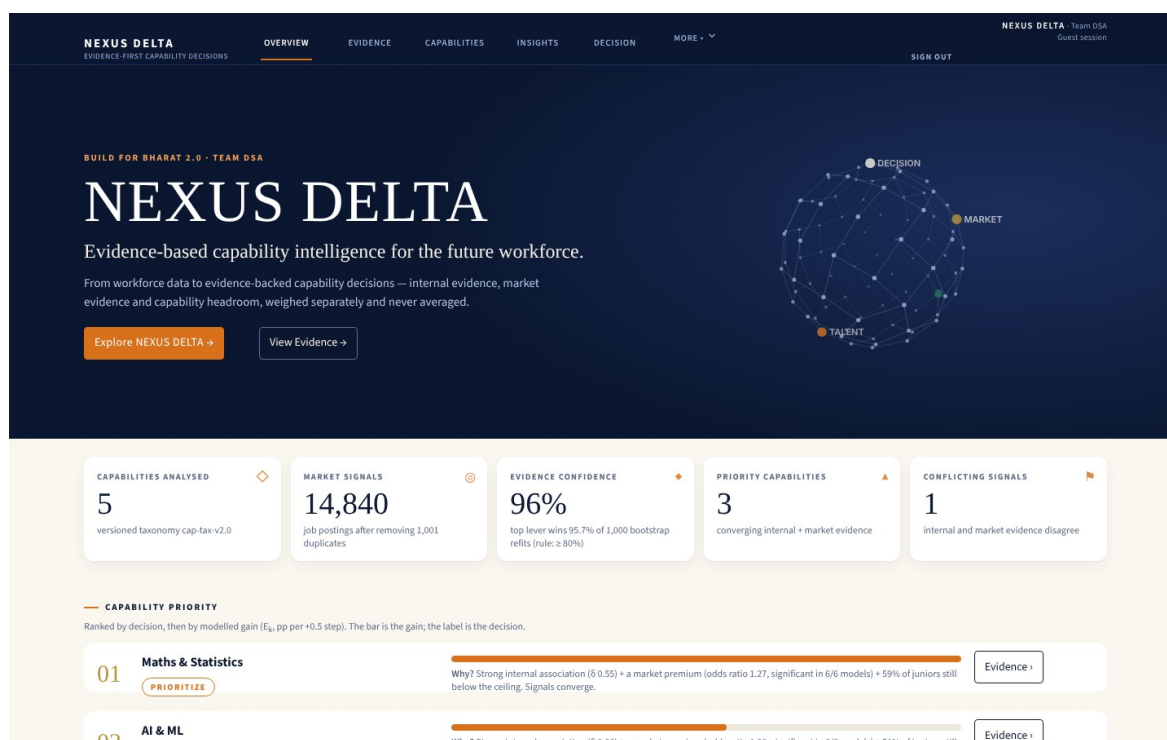


Figure 13. NEXUS DELTA — overview: decision ranking with a ‘Why?’ for every capability.

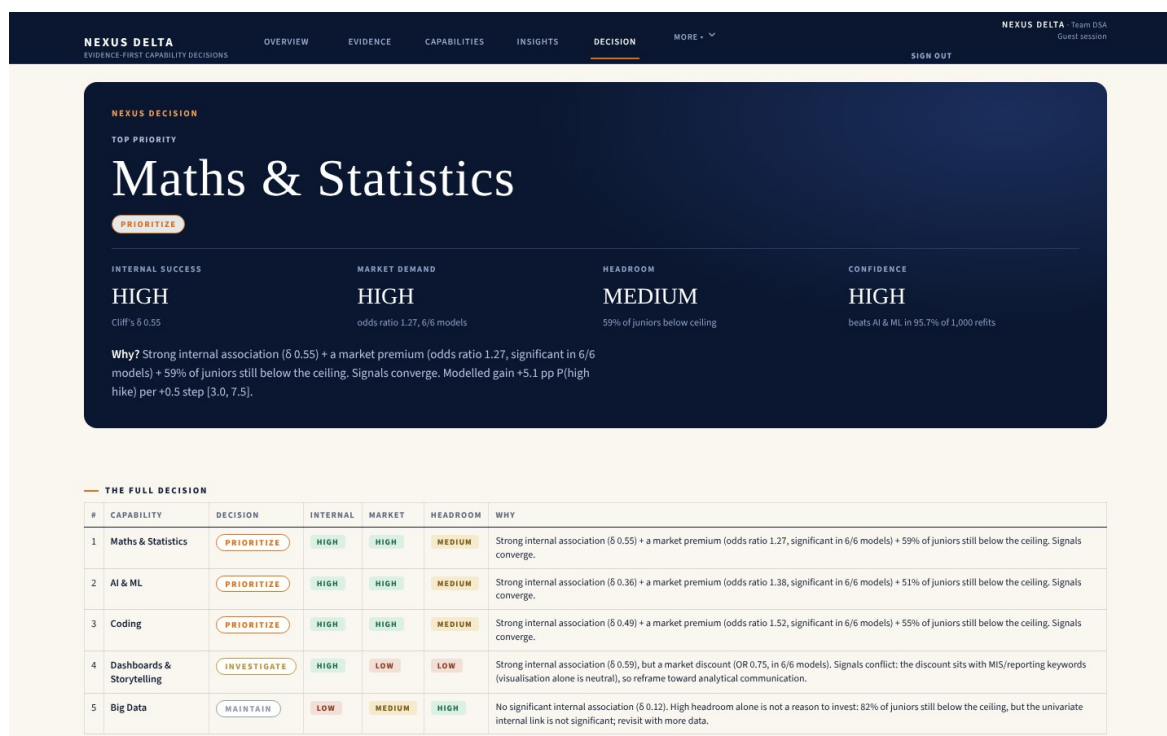


Figure 14. NEXUS DECISION screen — the same result as §5.1, with evidence levels and confidence.

6.3 Responsible use, built in

- **No fake joins, no fake salaries, no fake companies.** Simulated demo data is labelled **SIMULATED** on every screen.
- **Every output carries a label:** OBSERVED, DERIVED, INFERRED, PROPOSED or SIMULATED.
- **No automatic hiring or rejection.** Organisation views are discussion prompts; a human decides.
- **Privacy by design:** passwords hashed with scrypt; records encrypted with AES-256-GCM; employee culture answers stored without any user ID and shown only after 5 responses (a minimum group size).
- **Optional means optional:** the emotional-intelligence questionnaire can be skipped and never enters any score.

6.4 Limitations and next steps

Table 33. Limitations and next steps.

Limitation	Effect	Next step
JDS is one company, n = 139	Internal evidence may not generalise	Multi-company skills data; re-run NB02–03
All evidence is associational	Cannot claim a skill causes a hike	Pilot training cohort with a comparison group
87% of skills fields truncated	Under-detects skills	Full-text postings; hand-audited taxonomy sample
Salary only in bands	No exact pay modelling	Band distributions retained in product
Work DNA and culture survey not validated	Fit scores are indicative	Validate against retention with a partner employer
E _k computed in-sample	Optimistic gain estimates	Out-of-sample estimation with a larger cohort
Six specifications not jointly corrected	Some risk of selective reading	The rule needs a premium in both S5 and S6; all three premiums are in fact significant in all six

Limitation	Effect	Next step
SAS mirror not executed	SAS results not shown	Run in SAS VFL; compare with Python outputs

6.5 Closing

EVIDENCE, NOT OPINION

We started with a zero: no shared IDs, so no shortcuts.

We end with one decision, stated with its confidence: **Maths & Statistics — PRIORITIZE — first among funded capabilities in 91.3% of refits.**

Know what to build. Know what to learn. Know where you can thrive.

Appendix

A. Code excerpts

A.1 Internal evidence (NB02)

```
def cliffs_delta(hi, lo):
    U, p = mannwhitneyu(hi, lo, alternative="two-sided")
    return 2*U/(len(hi)*len(lo)) - 1, p

for k in SK:
    hi = JDS.loc[JDS.high_hike == 1, k].values
    lo = JDS.loc[JDS.high_hike == 0, k].values
    d, p = cliffs_delta(hi, lo)
    boots = [cliffs_delta(rng.choice(hi, len(hi)), rng.choice(lo, len(lo)))[0]
              for _ in range(2000)]
    rows.append(dict(skill=k, delta=d, p_raw=p,
                    ci_lo=np.percentile(boots, 2.5), ci_hi=np.percentile(boots, 97.5)))
eff["p_holm"] = multipletests(eff.p_raw, method="holm")[1]
```

A.2 Market evidence (NB04)

```
from statsmodels.miscmodels.ordinal_model import OrderedModel

def fit_or(df, suffix, spec, flags=FL):
    X = design(df, suffix, spec, flags)          # flags + nested controls S1..S5
    y = df.salary_ord.values                     # 6 ordered salary bands
    m = OrderedModel(y, X.values, distr="logit").fit(method="bfgs", maxiter=800, disp=False)
    ci = m.conf_int()[len(flags):]
    return {f: dict(OR=np.exp(m.params[i]), lo=np.exp(ci[i, 0]), hi=np.exp(ci[i, 1]))
                  for i, f in enumerate(flags)}

prem = {f: sum(out[s][f]["lo"] > 1 for s in out) for f in FL} # specs with a premium
```

A.3 Decision rule (NB06, condensed)

```
if d >= 0.33 and lo > 0:          internal = "Strong"
elif p_holm >= 0.05 or d < 0.147: internal = "Weak"
else:                             internal = "Moderate"

if internal == "Strong" and market in ("Premium", "Neutral"): verdict = "FUND"
elif internal == "Strong" and market == "Discount":           verdict = "FUND WITH REFRAME"
elif internal == "Moderate" and market == "Premium":          verdict = "MONITOR"
else:                                                           verdict = "DEPRIORITISE"

p_beat = float((B[:, lead] > B[:, runner_up]).mean()) # 0.957 over 1,000 refits
tie = p_beat < 0.80
```

A.4 Privacy gate (application, culture.py)

```
def aggregate(org_id, cyc=None):
    n = count(org_id, cyc)
    if n < K_MIN:
        # nothing is shown below 5 responses
        return dict(suppressed=True, n=n, need=K_MIN, cycle=cyc)
    rs = _load(org_id, cyc) # AES-256-GCM decrypt; rows carry no user id
```

B. Full market table (odds ratio [95% CI])

Table 34. Proportional-odds ratios for a higher salary band, all specifications (NB04).

Spec	Maths & Statistics	AI & ML	Coding	Dashboards & Storytelling	Big Data
S1 flags only	1.60 [1.42, 1.80]	1.93 [1.69, 2.21]	1.54 [1.38, 1.71]	0.69 [0.61, 0.78]	1.77 [1.51, 2.07]
S2 +experience	1.41 [1.24, 1.60]	2.08 [1.80, 2.40]	1.52 [1.36, 1.71]	0.75 [0.66, 0.85]	1.16 [0.99, 1.37]
S3 +seniority	1.31 [1.16, 1.49]	2.13 [1.85, 2.46]	1.53 [1.37, 1.71]	0.74 [0.65, 0.84]	1.16 [0.98, 1.36]
S4 +location	1.30 [1.14, 1.48]	2.13 [1.84, 2.45]	1.53 [1.36, 1.71]	0.74 [0.65, 0.83]	1.16 [0.98, 1.37]

Spec	Maths & Statistics	AI & ML	Coding	Dashboards & Storytelling	Big Data
S5 +all controls (primary tax.)	1.27 [1.11, 1.45]	1.38 [1.08, 1.75]	1.52 [1.35, 1.70]	0.75 [0.67, 0.86]	1.09 [0.89, 1.34]
S6 all controls (wide tax. & population)	1.42 [1.28, 1.57]	1.25 [1.04, 1.50]	1.55 [1.42, 1.70]	0.80 [0.73, 0.88]	1.15 [0.97, 1.36]

C. Application screens



Figure 15. Insights page: the live conflict matrix.

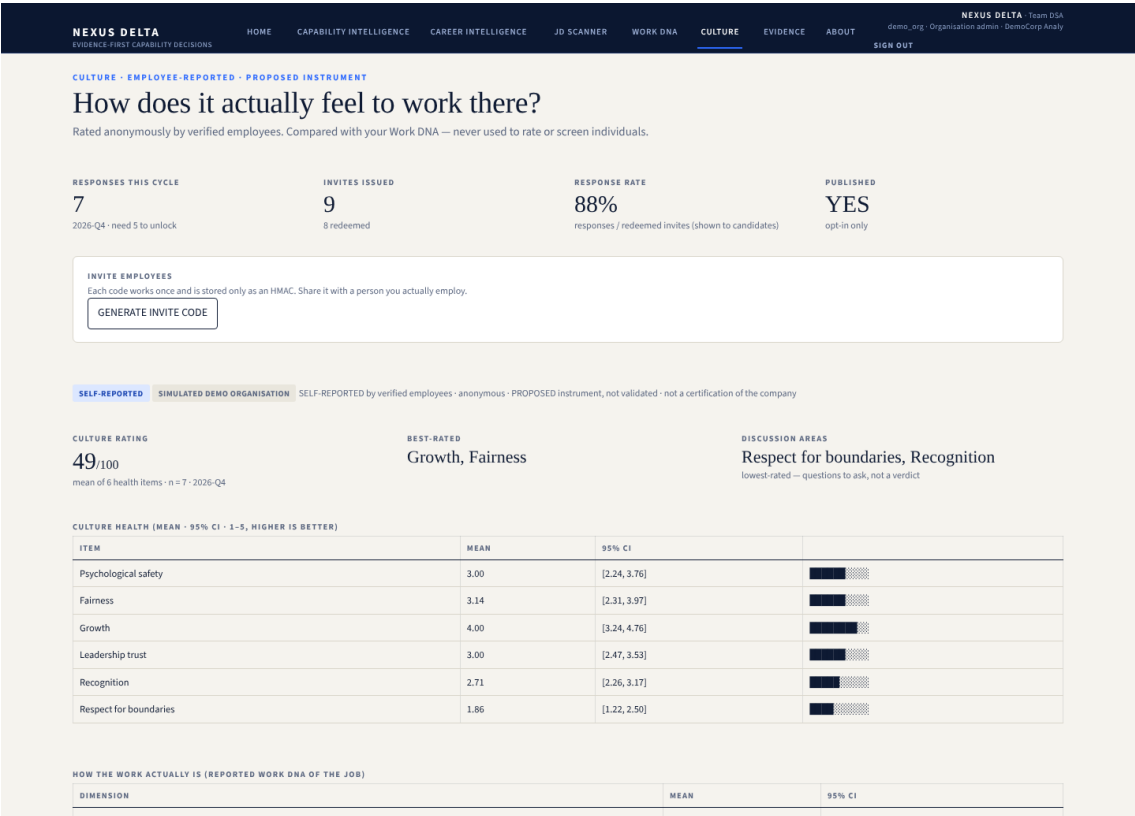


Figure 16. Culture page (organisation view, SIMULATED demo organisation): response count, invite codes and the scorecard, which appears only after 5 anonymous responses.

D. SAS mirror (excerpt, ready for SAS VFL; not executed)

nexus_delta_sas_mirror.sas (sections 1–3)

```

/* NEXUS DELTA - SAS mirror of the key analytical steps (Team DSA, Build For Bharat 2.0)
STATUS: translation of the Python pipeline for SAS Viya / SAS VFL /
SAS OnDemand for Academics.
Not executed in the authoring environment: re-run in SAS before quoting
any SAS-side number.
Numbers in the Approach Note come from the executed Python notebooks 01-08. */
options validvarname=v7; /* 'maths-stats_skills' -> maths_stats_skills */
/* folder holding the organiser files and the exported out/analytics_clean.csv */
libname nd "/home/&sysuserid/nexus_delta";

proc import datafile="/home/&sysuserid/nexus_delta/JDS_Skill_Traits.xlsx"
    out=nd.jds dbms=xlsx replace; run;
proc import datafile="/home/&sysuserid/nexus_delta/out/analytics_clean.csv"
    out=nd.analytics_raw dbms=csv replace;
    guessingrows=max; run;

/* three flags are exported as the text True/False; capability flags are already 0/1 */
data nd.analytics;
    set nd.analytics_raw;
    array t{*} $ is_ds_p jobtype_missing trunc_skills;
    array n{*} is_ds_p_n jobtype_missing_n trunc_skills_n;
    do i = 1 to dim(t); n{i} = (strip(t{i}) = 'True'); end;
    drop i;
run;

/* 1. Internal evidence: Mann-Whitney U (Wilcoxon rank-sum) per skill, high vs low hike */
%let skills = big_data_skills maths_stats_skills coding_skills ai_and_ml_skills
    dashboard_and_storytelling_skills;
proc nparlway data=nd.jds wilcoxon;
    class salary_hike_high_or_low;
    var &skills;
    ods output WilcoxonTest=nd.wtest WilcoxonScores=nd.wscores;
run;
/* two-sided p-values -> Holm adjustment (multtest needs a variable named raw_p) */
data nd.pvals;
    set nd.wtest; where Name1 = 'P2_WIL'; /* normal approx., continuity-corrected */
    raw_p = nValue1; keep Variable raw_p;
run;
proc multtest inpvalues=nd.pvals holm out=nd.holm_out; run;
/* Cliff's delta = 2U/(n1*n0) - 1, with U = rank sum of the high group - n1(n1+1)/2 */
proc sort data=nd.wscores; by Variable; run;
data nd.cliffs;
    merge nd.wscores(where=(Class='1') rename=(N=n1 SumOfScores=S1))
          nd.wscores(where=(Class='0') keep=Variable Class N rename=(N=n0));
    by Variable;
    U = S1 - n1*(n1+1)/2; cliffs_delta = 2*U/(n1*n0) - 1;
run;

/* 2. T1: logistic model on standardised skills + ROC */
proc stdize data=nd.jds out=nd.jds_z method=std; var &skills; run;
proc logistic data=nd.jds_z plots(only)=roc;
    model salary_hike_high_or_low(event='1') = &skills / ctable;
run;

/* 3. Market: cumulative-logit (proportional-odds) model of the 6 salary bands.
DESCENDING models P(Y >= j), so exp(beta) > 1 means higher bands. The score test
for the proportional-odds assumption is printed by default. */
proc logistic data=nd.analytics(where=(is_ds_p_n=1)) descending;
    class seniority(ref='mid') loc_tier(ref='Tier1') role_family(ref='other') / param=ref;
    model salary_ord = big_data_p maths_stats_p coding_p ai_ml_p story_p
        exp_min exp_span jobtype_missing_n trunc_skills_n
        seniority loc_tier role_family / link=clogit;
    ods output OddsRatios=nd.market_or;
run;

```

E. Reproduce everything

Notebook	Open
01_data_exploration_cleaning	view on GitHub

Notebook	Open
02_internal_evidence_JDS_SDS	view on GitHub
03_T1_high_hike_model	view on GitHub
04_market_salary_band_model	view on GitHub
05_T2_JD_scanner_salary_band	view on GitHub
06_headroom_career_decision_engine	view on GitHub
07_app_role_profiles	view on GitHub
08_supplementary_checks	view on GitHub
Application	streamlit run app.py (offline) — repository

F. Glossary

Term	Meaning
Cliff's δ	Probability a high-group score exceeds a low-group score minus the reverse; -1 to 1
Odds ratio (OR)	Multiplicative change in the odds of a higher salary band when a capability is present; 1 = no effect
Headroom	Share of juniors below the 5.0 ceiling in a capability
E _k	Modelled change in P(high hike) for a +0.5 improvement in capability k (what-if)
Quadratic κ	Agreement for ordered categories; penalises far misses more
Minimum group size (K = 5)	No group statistic shown until at least 5 people contributed
PRIORITIZE / INVESTIGATE / MAINTAIN	Product labels for the notebook verdicts FUND / FUND WITH REFRAME, MONITOR or TIED / DEPRIORITISE